Project Parcel — Project Report

1. PROBLEM AS UNDERSTOOD, INCLUDING ASSUMPTIONS

The task is to identify shipments likely to arrive after their promised delivery date early enough for an operations team to consider an intervention. A shipment is labeled late when actual_delivery_date is later than promised_delivery_date. This is a clear, auditable definition, although a grace period or business-hours definition could produce different labels.

2. APPROACH AND KEY DECISIONS

I audited missing values, duplicate shipment IDs, reference-table joins, and category formatting. There were no duplicate shipment IDs and no unmatched carrier or warehouse joins in the recorded audit. Missing values included distance (200), package weight (144), declared value (160), and destination region (96). Missingness was retained for preprocessing rather than silently deleting shipments. Category strings were normalized to reduce duplicate-looking categories.

3. RESULTS, WITH HONEST UNCERTAINTY

On the held-out chronological test set, the recorded results were approximately: PR-AUC 0.5468, ROC-AUC 0.6827, top-10% precision 65.83%, top-10% recall 18.33%, F1 0.2868, and Brier score 0.2246. The top-10% policy selects 120 of 1,200 test shipments. About two thirds of the selected shipments were late, but the policy captured fewer than one fifth of all late shipments.

4. WHAT FAILED AND WHAT WAS LEARNED

During evaluation, scikit-learn raised a TypeError because pandas nullable values (`pd.NA`) reached its missing-value checks. The fix was to normalize nullable feature dtypes consistently across training, validation, and test data, replacing nullable missing values with NumPy-compatible missing values where appropriate. I learned that data types are part of the model interface and must be tested across every split.

5. RISKS, LIMITATIONS, AND WHAT I WOULD DO WITH TWO MORE WEEKS

The model misses many late shipments at the chosen review capacity. It should support prioritization, not replace operational judgment or be described as a complete detector. Synthetic data may not represent real carrier behavior, seasonality, operational disruptions, or data quality. Feature leakage remains a risk unless all inputs are proven available at dispatch. Drift checks require ongoing monitoring and labeled outcomes to detect performance decay, not just feature-distribution changes.

6. HOW TO RUN EVERYTHING

Open the final notebook in Google Colab, follow its setup and synthetic-data preparation cells, and run all cells from a clean runtime in order. The notebook saves the fitted model and runs evaluation, sample predictions, and drift/smoke checks. For the helper package, install requirements.txt and run `python -m pytest -q`. See README.md for the repository layout, assumptions, and limitations.